

Motivation Letter: Software Engineer for HPC+ML Applications (RE3)

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Recruitment Panel

Barcelona Supercomputing Center - Centro Nacional de Supercomputación

Subject: Application for Software Engineer for HPC+ML Applications (RE3), reference 90_26_CASE_WP_RE3

Dear Members of the Recruitment Panel,

I am applying for the Software Engineer for HPC+ML Applications position because it brings together three areas that have shaped my development: physical modelling, machine learning, and reliable software workflows. I hold a BSc in Mechanical Engineering and am completing an MSc in Modelling for Science and Engineering at the Universitat Autònoma de Barcelona, specialising in Data Science. My professional and research work now focuses on structured AI systems, while my earlier training and engineering experience remain grounded in physical models and numerical analysis.

My BSc thesis built and validated a neural-network model for predicting turbulent intensity from wind-profile data. The project required connecting a physical problem with data preparation, model design, validation, and interpretation rather than treating machine learning as an isolated exercise. During my engineering degree and later at ABS Consulting, I also worked with ANSYS Fluent, Abaqus, Mathcad, Inventor, and STAAD.Pro. At ABS, I supported seismic qualification and structural assessments for UK nuclear power assets, where reviewable calculations, standards compliance, and technical reporting were essential.

At Aily Labs, I have worked from prompt-defined MCP capabilities and gated skills through semantic SQL-agent access and production evaluation. Approved table descriptions and usage guidance provide bounded access to enterprise data; package-owned question sets, deterministic validators, traces, and LLM-as-a-judge scoring then measure whether the agent used the right data and tools. Working across the Super Agent, MCP and data services has strengthened my Python engineering, CI/CD, debugging, and ability to attribute failures across a distributed application.

Through a JAE Intro ICU 2025 scholarship at IIIA-CSIC, I also work on a ROS 2 architecture connecting dialogue, symbolic world state, language-model planning, deterministic execution, and trace-based evaluation for the NAO robot. The stack has reinforced the same scientific-software principle: results need typed inputs, observable execution, and evidence that separates a model error from an integration or measurement error.

My principal personal ML project, iTrader, is an RLVR-style trading research platform whose implemented baseline combines market-window simulation, validated task curricula, an RLlib PPO/GAE training loop, and task-conditioned FiLM policy and value networks. Its reward registry keeps PnL, drawdown and turnover components finite, decomposed, normalised across relevant scales, and separately observable. Local LLM proposer backends and deterministic fallbacks generate task candidates, while run manifests retain code, configuration, seed, and dataset identity. The intended route to live trading adds capability-profiled solver heads, grouped proposer optimisation, Market-JEPA representations, and replay or shadow evaluation before deterministic risk and execution.

The Watson Stack integrates local llama.cpp serving, LiteLLM routing, and tool-based agent workflows. My unpublished Universal Agentic Harness H0 proof studies typed task and capability contracts, portable adapters, deterministic evidence gates, and replayable traces. These projects are not substitutes for production HPC experience, but they demonstrate how I approach computational systems: explicit interfaces, bounded experiments, observable execution, and documentation that supports reproducible iteration.

The Wave Phenomena role is an ambitious application for my current level. I do not yet claim expert C/C++ programming, leadership of European projects, or three to five years in a directly comparable HPC position. My experience with large-scale HPC facilities, GPU porting, and distributed scientific software is also developing rather than established. I am applying because my combination of physical modelling, machine learning, Python systems work, and rigorous engineering practice offers a relevant foundation, and because the Earth-AID and geoscience focus is a compelling setting in which to deepen those capabilities.

I would bring curiosity, disciplined validation, clear communication, and experience moving between domain experts and software abstractions. BSC's ability to unite scientific research, modern computational infrastructure, and real engineering applications makes it an exceptional environment for that next step.

Thank you for considering my application. I would welcome a discussion about whether my interdisciplinary background and growth trajectory fit the needs of the Wave Phenomena group.

Yours sincerely,

Juan David Bendek Williamson

References

1. **Roger Montane**, Mid Data Scientist, Aily Labs

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2. **Dr. Balogh Miklos**, Professor, BME

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